SIH 2026 · PS 26052 (DRDO) · Team ZEIT v.1.0 (HW2026106)

ZEIT Hardware

Every part, every port, every wire, and how the headset is built

Version	2.0 — draft for team review
Date	17 September 2026
Status	Designed, not built. Every claim carries a status tag
Companion document	ZEIT System Architecture (step IDs A1–A11, R1–R5, B1–B11, T1–T8)
Supersedes	Architecture v1 (Raspberry Pi 5 design) in handbook/index.html and docs/01-system/architecture.md

Five blank revision pages are provided at the end of this document for your changes.

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1. How to use this document

Each part has an ID (H1–H24). The "Used by" column names the architecture steps. If a part changes, check those steps; if a step changes, check the parts it names. Prices are September 2026 estimates including GST; confirm with quotations.

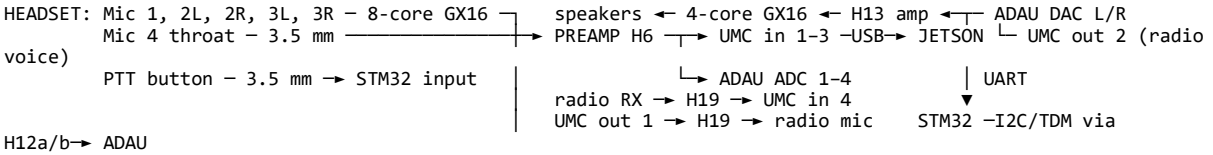
Status tags

Tag	Meaning
VERIFIED	Checked against a published source (datasheet, standard, manufacturer page)
MEASURED	We ran it ourselves and have the number
ESTIMATE	Engineering arithmetic; not yet tested
UNVERIFIED	A risk. Must be checked before relying on it

2. Summary

> **IN PLAIN WORDS** A modified 3M PELTOR earmuff with six microphones and two speakers. One ADAU1787 DSP protects both ears; a Jetson runs the AI that cleans both voices; a walkie-talkie pair stands in for the military radio.

Headset	3M PELTOR Optime III, modified: 6 mics, 2 speakers
Lane A	EVAL-ADAU1787Z (one board, both ears) + STM32 Nucleo-H743ZI2
Lane B	Jetson Orin Nano Super + Behringer UMC404HD
Radio	PMR446 pair (WPC ETA), isolated interface
Must-buy total	≈ ₹1,11,550 – ₹1,37,250 (ESTIMATE)
Budget requested	₹1,25,000



3. Bill of materials (H1–H24)

ID	Part	Key spec	Qty	Est. ₹	Used by	Buy
Headset and sensors						
H1	3M PELTOR Optime III H540A earmuff (alt. X5A)	Passive, SNR 35 dB (X5A 37) — VERIFIED; double casing	1	2,000–5,000	Protection; shell	Must
H1-P	Basic earmuff	Practice	1	~500	Fitting practice	Must
H1-A	40 mm drivers, 32 Ω	2 fitted + 2 spare	4	400–1,000	Speakers (A10–A11, R5)	Must
H1-B	Gooseneck electret boom mic	2-wire electret	1	300–800	Mic 1 (B1, B7)	Must
H1-C	Acrylic / plastic sheet	Speaker rings	1	150–300	Speaker mounts	Must
H1-D	Acoustic foam sheet	Replaces cup foam	1	200–400	Damping, seal	Must
H1-E	Boom clamp	Left cup	1	100–300	Mic 1	Must
H1-F	Acoustic mesh	Covers inside mics	1	50–150	Mic 3L, 3R	Must
H1-G	PTT button + 3.5 mm cable	Momentary	1	100–300	B11	Must
H2	Infineon IM73A135 breakout	AOP 135 dB, SNR 73 dB, VDD 2.3–3.0 V — VERIFIED	2	1,600–5,000	Mic 2L, 2R (A1–A6, B1–B7)	Must (Gate 3)
H3	SparkFun ICS-40180 MEMS breakout (BOB-18011)	1.5–3.6 V, SNR 65, op-amp output — VERIFIED; check stock	3	750–1,500	Mic 3L, 3R + spare	Must
H4	Throat mic, 3.5 mm	Electret laryngophone	1	1,500–3,000	Mic 4 (B1, B7, B9)	Must
H5	Foam windshields	Mic foams	3	300–700	Mic 2L, 2R, 1	Must
H6	Preamp / bias board	6 in, 7 out, L/R switch	1	3,000–3,500	All mics	Must (build)
H7	Foam earplugs	Rated	1 box	~300	Double protection	Must
H8	Cables, GX16, sealing	8-core + 4-core shielded, GX16-8 + GX16-4, grommets, silicone	1 set	1,300–2,200	Headset connections	Must
Lane A electronics						
H9	EVAL-ADAU1787Z	4 ADC, 2 DAC, FastDSP 768 kHz — VERIFIED	1	~32,000	A1–A11	Must (Gates 1, 2)
H10	USBi programmer	SigmaStudio interface	0–1	0 / ~9,000	Programming H9	Only if needed

ID	Part	Key spec	Qty	Est. ₹	Used by	Buy
H11	STM32 Nucleo-H743ZI2	UART, I²C, SAI, ADC, button	1	~3,500	Boot, modes, A8, PTT, volume	Must
H12a	PCA9306 I²C level shifter	1.8 ↔ 3.3 V, made for I²C	1	100–200	H11 ↔ H9 control	Must unless on-board
H12b	TXB0104 push-pull shifter	1.8 → 3.3 V, BCLK / FSYNC / SDATA	1	100–200	H9 → H11 (A8)	Phase 2
H13	TPA6130A2 amp	2.5–5.5 V, run at 3.3 V; 32 Ω; I²C volume	1	1,000–2,000	Sums DSP + radio; drives speakers	Must
Lane B electronics						
H14	Behringer UMC404HD	4 in, 24-bit, USB class-compliant	1	~15,000	B1, B11, R1, R5	Must
H15	Jetson Orin Nano Super Dev Kit	6× A78AE, Ampere 1024 CUDA, 8 GB, 67 TOPS, 7–25 W — VERIFIED	1	37,200–45,200	B1–B11, R1–R5	Must (Gate 6)
H16	microSD 128 GB A2	Boot + storage	1	~1,500	H15	Must
H17	Hum fix	Grounding → ground-loop isolators → high-speed USB isolator (not ADuM3160)	0–1	0 / check	B1	Only if hum
Communication unit						
H18	PMR446 pair	WPC ETA; K-type jack — rule VERIFIED	1 pair	2,500–6,000	B11, R1	Must (Gate 5)
H19	Radio interface parts	2 transformers, PC817, resistors, K-plug cable	1 set	600–1,200	B11, R1	Must
Power and test						
H20	Clean 5 V supply	Separate from Jetson	1	~1,000	H6, H9, H11 (H13 from the Nucleo 3.3 V)	Must
H21	PD bank + 19 V trigger	Field power	0–1	0 / ~5,000	H15	Outdoor demo only
H22	Mannequin + ear mic	Headset must fit	1	~2,000	Calibration, Lane A tests	Must
H23	Sound level meter	UNI-T UT353BT class	1	~2,500	Calibration, safety	Must
H24	Calibrator 94 dB / 1 kHz	—	0–1	0 / 6,000–12,000	Absolute dB SPL	Optional

Group	Est. ₹
Headset and sensors	12,550 – 24,950
Lane A electronics	36,700 – 37,900
Lane B electronics	53,700 – 61,700
Communication unit	3,100 – 7,200
Power and test	5,500
Must-buy total	1,11,550 – 1,37,250
Conditional (H10, H17, H21)	up to ~14,000 + isolator if needed
Optional (H24)	6,000 – 12,000

Tools if the lab lacks them: mini drill + step bit, temperature-controlled soldering iron, multimeter, glue gun (≈ ₹2,600–4,800).

4. Microphone map

Mic	Position	Part	Preamp in	Preamp out	Destination
Mic 1	Boom, left	H1-B	IN1	OUT1	UMC in 1 → AI input 0
Mic 2L	Outside left	H2	IN2	OUT5 (+ OUT2 if switch = L)	ADAU ADC 1
Mic 2R	Outside right	H2	IN3	OUT6 (+ OUT2 if switch = R)	ADAU ADC 2 + UMC in 2 → AI input 1
Mic 3L	Inside left	H3	IN4	OUT7	ADAU ADC 3
Mic 3R	Inside right	H3	IN5	OUT8	ADAU ADC 4
Mic 4	Throat	H4	IN6	OUT3	UMC in 3 → AI input 2

5. Preamp / bias board (H6)

Function	Design
Inputs	6: IN1 boom (electret), IN2–IN5 MEMS (3-wire), IN6 throat (electret)
Bias	Electret 2–5 V via resistor (IN1, IN6); MEMS on a shared 2.8 V LDO (IM73A135 max 3.0 V — never 3.3 V)
Gain	Per channel, OPA1678 on 5 V (2.5 V mid-rail, ≈ 1.2 Vrms max); a shout ≈ −12 dBFS at the converters
ADAU side	AC-coupled, 1 kΩ series, gain below the ADAU full scale (value from the datasheet)
UMC side	TRS line inputs, phantom power OFF
Outputs	7 buffered: OUT1–OUT3 → UMC in 1–3; OUT5–OUT8 → ADAU ADC 1–4 (OUT4 unused)
Handedness switch	SPDT: Mic 2R (right-handed, default) or Mic 2L (left-handed) to OUT2
Supply	Clean 5 V (H20); star ground on this board

6. Board port assignments

EVAL-ADAU1787Z (H9)

Port	Signal	Notes
ADC 1–4	Mic 2L, Mic 2R, Mic 3L, Mic 3R	From H6 OUT5–OUT8
DAC L / R	→ H13 (summing resistors)	79 dB share
I²C	← H11 via H12a (PCA9306)	Control, program load
Serial (I²S/TDM)	→ H11 SAI via H12b (TXB0104)	Adaptive ANC phase only
USB	Laptop, SigmaStudio	Development; power

Behringer UMC404HD (H14)

Port	Signal	Notes
In 1 / 2 / 3 (TRS)	Mic 1 / reference / Mic 4	AI inputs 0 / 1 / 2; gains set once and logged
In 4 (TRS)	Radio RX (field) · mannequin ear mic (lab)	Swap the cable
Out 1	→ H19 → radio mic in	Outgoing voice
Out 2	→ H13 (summing resistors)	Incoming voice
48 V phantom	OFF	H6 powers the mics

Jetson Orin Nano Super (H15)

Port	Use	Notes
USB	UMC404HD	Audio class
40-pin UART	STM32 link	3.3 V; assign pins from NVIDIA's header pinout
40-pin GPIO	Not used for PTT	Header pins go through TXB0108 shifters (≈ 2–3 mA) — cannot drive the optocoupler
DisplayPort	Monitor	Not HDMI
microSD / DC	H16 / 19 V adapter (H21 in field)	—

STM32 Nucleo-H743ZI2 (H11)

Port	Use
UART	Jetson link (protocol in the architecture document)
I²C	ADAU1787 (via H12a) and TPA6130A2 volume
SAI	Mic data from the ADAU for A8 (via H12b)
GPIO in / out	PTT button (pull-up) / PC817 LED via 330 Ω (≈ 6 mA)

Port	Use
3V3 pin	Supplies H13
User button	ANC on / PASSIVE
USB	Power (H20) and flashing

Radio interface (H19)

Path	Parts	Notes
UMC out 1 → radio mic	Divider → transformer 1 → 1–4.7 µF DC-block cap	The radio puts bias on its mic line and keys PTT when it is grounded; the cap stops the transformer shorting it
STM32 GPIO → radio PTT	PC817 optocoupler	Isolated; check which K-plug contact keys PTT
Radio speaker → UMC in 4	Divider → transformer 2	Isolated; both primary ends float (speaker amp may be bridged)

7. Cables and connector pinouts

Microphone cable — 8-core shielded, GX16 8-pin

Pin	Signal
1	MEMS supply (VDD)
2	Microphone ground
3	Mic 1 (electret signal + bias)
4	Mic 2L
5	Mic 2R
6	Mic 3L
7	Mic 3R
8	Shield (grounded at the preamp end only)

Speaker cable — 4-core shielded, GX16 4-pin

Pin	Signal
1	Left +
2	Left −
3	Right +
4	Right −

Mics and speakers in separate cables so speaker current cannot couple into the mics. Mic 4 and the PTT button have their own 3.5 mm leads. Label every wire at both ends.

8. Power and grounding

Board	Supply	Why
Jetson (H15)	Own 19 V adapter; H21 in the field	Keeps switching noise from the mics
UMC404HD (H14)	USB bus power from the Jetson	Phantom off keeps current low
H6, H9, H11, H13	Separate clean 5 V (H20)	Mic noise floor depends on it
Radio (H18)	Own batteries	Isolated by H19

- Star ground at the preamp board; shields grounded at one end; short leads.
- Hum: fix grounding first; then ground-loop isolators on analog lines; a high-speed USB isolator last. ADuM3160 boards are full-speed only and will not carry the UMC404HD.

9. Building the headset

> **IN PLAIN WORDS** Buy a ready earmuff and add the parts: speakers inside both cups, an inside mic next to each ear, an outside mic on each cup, a boom mic on the left. The throat mic is worn separately on the neck. Practise every drilling step on the cheap earmuff first.

1. **Open the cups.** Remove cushions and inner foam; note how they seal.
2. **Speakers (H1-A, H1-C).** Ring per cup holding a 40 mm driver facing the ear; seal with silicone. The cup must stay airtight — that is the passive protection.
3. **Inside mics (H3, H1-F).** One per cup, between speaker and ear, close to the ear canal, facing the ear, on a foam pad; mesh over. Do not touch the speaker cone.
4. **Outside mics (H2, H5).** One small hole through both casing layers per cup (step bit, slow, masking tape). Board facing out, recessed; seal both layers; windshield. Mic 2R on the right cup (away from the boom), Mic 2L on the left.
5. **Boom (H1-B, H1-E, H5).** Clamp the gooseneck to the left cup; tip 2–5 cm from the corner of the mouth; windshield.
6. **Wiring (H8).** One grommeted exit per cup; left-cup wires along the headband; 8-core mics, 4-core speakers, GX16 per section 7; strain relief.
7. **Throat mic (H4), PTT (H1-G).** On their own leads.
8. **Close up.** New foam (H1-D), cushions back; check the seal on a head.

10. Tests after building

Test	How	Pass
Wiring	Multimeter end to end and between adjacent pins	Continuous, no shorts
Every mic alive	Record each channel while tapping near it	Each mic on its own channel
Speakers	Slow sweep into each ear	Clean, no rattle, L/R separate
Seal	Loud noise outside; outside vs inside mic, system off	Inside clearly quieter, consistent with the rating
Voice leak	Speak in a quiet room; Mic 1 vs reference	Reference ≥ 25 dB quieter; otherwise move Mic 2R further back
Throat mic	Speak with loud noise playing	Voice clear, background low

Then calibration (architecture document, section 6).

11. Hardware ↔ neural network compatibility

Link	Hardware	Model / training	Status
Inputs and order	UMC in 1 / 2 / 3	Inputs 0 / 1 / 2	Locked; label cables
Sample rate	48 kHz	16 kHz	Resampler MEASURED
Frame / hop	768 @ 48 kHz	256 @ 16 kHz	Parity MEASURED
Reference overload	Clips above 135 dB	Clipping augmentation	Built
Gains	H6 per channel	Random per-channel gain	Input 1 built; input 2 to add
Voice in reference	Head between boom and Mic 2R	Leak −45 … −25 dB	Built; voice-leak test
Throat channel	H4	Real + simulated + dropout	To implement
Handedness	Switch on H6	Same model either way	No retraining
TensorRT	H15	GRU layers run in FP32 / FP16, not INT8 (VERIFIED, NVIDIA docs)	GRUs stay FP16; INT8 only for convolutions; engine build on the Jetson still to do

AUDIT A link-by-link compatibility audit (docs/01-system/compatibility-audit.md) found and fixed: PTT moved from the Jetson to the STM32; PCA9306 + TXB0104 instead of one 8-channel shifter; MEMS on 2.8 V (IM73A135 max 3.0 V); H3 named (ICS-40180); TPA6130A2 on 3.3 V; DC-block capacitor on the radio mic line; AC coupling into the 1.8 V ADAU; UART protocol defined; data loader and live runner moved to 3 inputs.

12. Budget

	Est. ₹
Must-buy total	1,11,550 – 1,37,250
Budget requested	1,25,000
What fixes the figure	Gate 1 (USBi), Gate 4 (earmuff price), Gate 6 (Jetson quote)

13. Gates and checklists

Before ordering

1. UG-1532: built-in USB? I/O voltage and level shifting? self-boot? serial pins? (H9–H12)
2. ADAU1787 FastDSP capacity for two ANC paths (H9)
3. IM73A135 as a ready breakout in India (else another ≥ 130 dB analog MEMS) (H2)
4. Earmuff price; cup depth (H1)
5. PMR446: WPC ETA, K-type jack, pinout (H18, H19)
6. Written Jetson quotation (H15)

After arrival

1. Record 4 channels from the UMC404HD on the Jetson (arecord).
2. Build the 3-input model as a TensorRT engine; frame time < 16 ms.
3. Build the headset; run the section 10 tests.
4. SigmaStudio: hear-through + limiters, then fixed ANC (both ears), then adaptation.
5. Calibrate on the mannequin.
6. Hum check; H17 only if needed.

14. Decisions log

Question	Decision	Why
Raspberry Pi or Jetson?	Jetson Orin Nano Super	PS names Jetson and TensorRT
Aviation headset or earmuff?	Earmuff, modified	No built-in electronics; more protection; cheaper
Bose A30?	No	Active ANC of its own; ≈ ₹1 lakh+
How many mics?	Six; three into the AI	Outside + inside per ear; voice, reference, throat for the AI
One or two ADAU1787?	One	4 in / 2 out fits stereo hybrid ANC once the radio joins at the amp
Process incoming radio voice?	Yes, on the Jetson	Intelligibility applies to what the wearer hears
AI reference mic?	Mic 2R (2L for left-handed)	Far side of the head from the mouth
Fixed or adaptive ANC first?	Fixed first	Lower risk
USB isolator?	Only if hum; not ADuM3160	ADuM3160 is full-speed only
PTT or VOX?	PTT	VOX would transmit gunshots

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